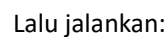


 Screenshot 1.1: Buka browser <http://localhost:8900>, screenshot dashboard HAProxy (kedua node hijau UP).



```
docker exec db-master mysql -h haproxy -P 3300 -u app_user -pAppPass123! -e "INSERT INTO
klinik_db.pasien (nama, tanggal_lahir, alamat, no_telepon) VALUES ('Pasien Uji HA', '1992-04-12', 'Jl.
Matraman No. 12', '081234567890');"
```

```
docker exec db-slave mysql -u root -pRootPass123! -e "SELECT id, nama, alamat FROM klinik_db.pasien WHERE nama='Pasien Uji HA';"
```

 Screenshot 1.2: Screenshot terminal yang menunjukkan INSERT berhasil dan data muncul di slave.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h hap
roxy -P 3300 -u app_user -pAppPass123! -e "INSERT INTO klinik_db.pasien (nama, t
anggal_lahir, alamat, no_telepon) VALUES ('Pasien Uji HA', '1992-04-12', 'Jl. Ma
traman No. 12', '081234567890');"
mysql: [Warning] Using a password on the command line interface can be insecure.

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -u root
-pRootPass123! -e "SELECT id, nama, alamat FROM klinik_db.pasien WHERE nama='Pa
sien Uji HA';"
mysql: [Warning] Using a password on the command line interface can be insecure.
id      nama      alamat
4       Pasien Uji HA   Jl. Matraman No. 12

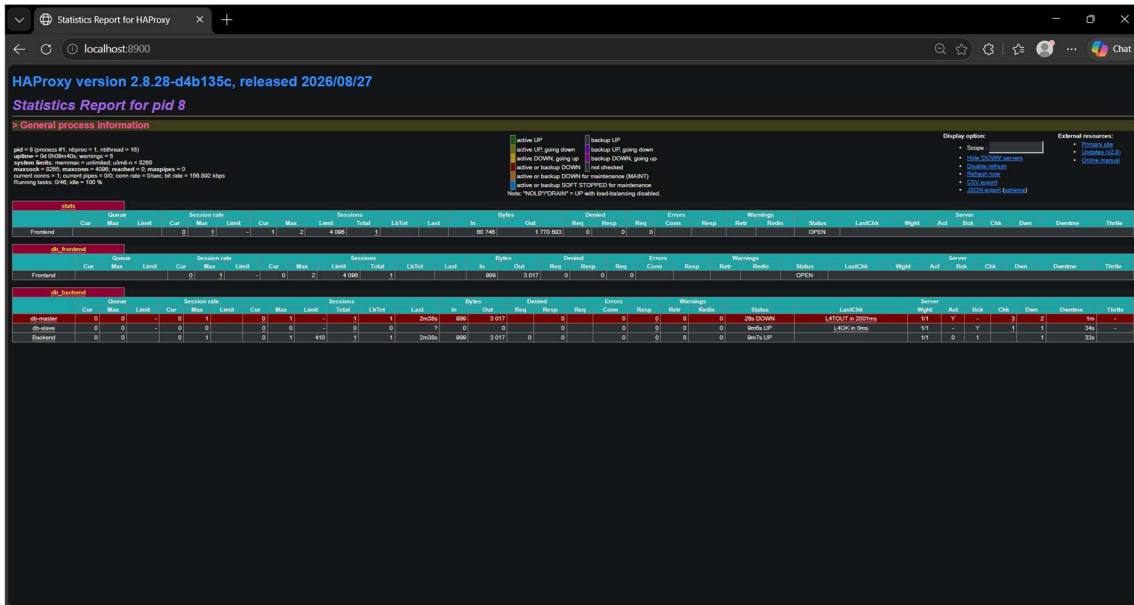
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Fase B: During-Attack (Matikan Master)

cmd

```
docker stop db-master
```

 Screenshot 1.3: Refresh browser <http://localhost:8900>, screenshot dashboard HAProxy (db-master MERAH/DOWN).



cmd

```
docker exec db-slave mysql -h haproxy -P 3300 -u app_user -pAppPass123! -e "SELECT id, nama FROM klinik_db.pasien LIMIT 3;"
```

cmd

```
docker exec db-slave mysql -h haproxy -P 3300 -u app_user -pAppPass123! -e "INSERT INTO klinik_db.pasien (nama) VALUES ('Data Saat Master Down');"
```

 **Screenshot 1.4:** Screenshot terminal — SELECT berhasil, INSERT ditolak dengan error --read-only.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker stop db-master
db-master

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -h hapr
oxy -P 3300 -u app_user -pAppPass123! -e "SELECT id, nama FROM klinik_db.pasien
LIMIT 3;"
mysql: [Warning] Using a password on the command line interface can be insecure.
id      nama
1       Ahmad Fauzi
2       Siti Rahayu
3       Budi Santoso

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -h hapr
oxy -P 3300 -u app_user -pAppPass123! -e "INSERT INTO klinik_db.pasien (nama) VA
LUES ('Data Saat Master Down');"
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1290 (HY000) at line 1: The MySQL server is running with the --read-only o
ption so it cannot execute this statement

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Fase C: After-Mitigation (Pulihkan Master)

cmd

docker start db-master

Tunggu ~10 detik, lalu:

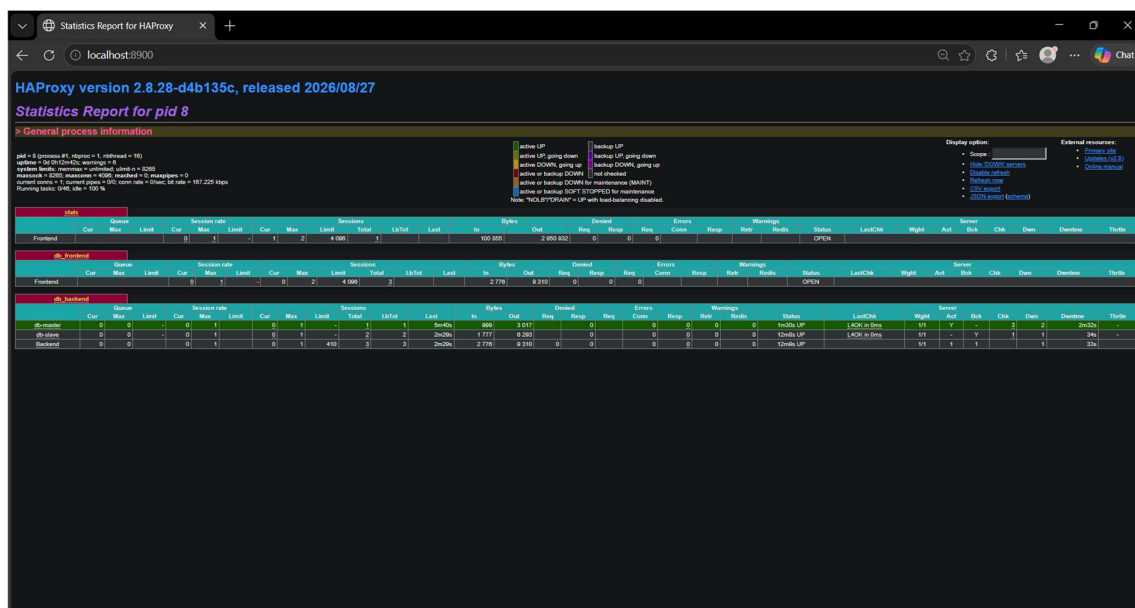
cmd

docker exec db-slave mysql -u root -pRootPass123! -e "SHOW REPLICA STATUS\G"

 **Screenshot 1.5:** Screenshot terminal — Replica_IO_Running: Yes dan Replica_SQL_Running: Yes.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -u root -pRootPass123! -e "SHOW REPLICA STATUS\G"
mysql: [Warning] Using a password on the command line interface can be insecure.
***** 1. row *****
      Replica_IO_State: Waiting for source to send event
        Source_Host: db-master
        Source_User: replicator
        Source_Port: 3306
        Connect_Retry: 60
        Source_Log_File: mysql-bin.000004
        Read_Source_Log_Pos: 26432
        Relay_Log_File: 83276e418d88-relay-bin.000004
        Relay_Log_Pos: 26648
        Relay_Source_Log_File: mysql-bin.000004
        Replica_IO_Running: Yes
        Replica_SQL_Running: Yes
        Replicate_Do_DB:
        Replicate_Ignore_DB:
        Replicate_Do_Table:
        Replicate_Ignore_Table:
        Replicate_Wild_Do_Table:
        Replicate_Wild_Ignore_Table:
          Last_Errno: 0
          Last_Error:
          Skip_Counter: 0
        Exec_Source_Log_Pos: 26432
        Relay_Log_Space: 27034
        Until_Condition: None
        Until_Log_File:
        Until_Log_Pos: 0
        Source_SSL_Allowed: No
        Source_SSL_CA_File:
        Source_SSL_CA_Path:
        Source_SSL_Cert:
        Source_SSL_Cipher:
        Source_SSL_Key:
        Seconds_Behind_Source: 0
        Source_SSL_Verify_Server_Cert: No
          Last_IO_Errno: 0
          Last_IO_Error:
          Last_SQL_Errno: 0
          Last_SQL_Error:
        Replicate_Ignore_Server_Ids:
```

 **Screenshot 1.6:** Refresh browser <http://localhost:8900>, screenshot dashboard HAProxy (kedua node kembali hijau).



SKENARIO 2 — SQL INJECTION ATTACK & PREVENTION

Fase A: Before-Attack (Baseline)

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "SELECT id, nama, tanggal_lahir FROM klinik_db.pasien;"
```

 **Screenshot 2.1:** Screenshot terminal — daftar data pasien normal (baseline).

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "SELECT id, nama, tanggal_lahir FROM klinik_db.pasien;"
mysql: [Warning] Using a password on the command line interface can be insecure.
id      nama      tanggal_lahir
1      Ahmad Fauzi    1990-05-15
2      Siti Rahayu    1985-08-22
3      Budi Santoso   1995-12-01
4      Pasien Uji HA  1992-04-12
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Fase B: During-Attack (Injeksi SQL — Data Bocor)

Serangan pada tabel pasien:

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "USE klinik_db; SET @input = '\" OR '1'='1' -- '; SET @q = CONCAT('SELECT id, nama FROM pasien WHERE nama = '", @input, "'); PREPARE s FROM @q; EXECUTE s; DEALLOCATE PREPARE s;"
```

 **Screenshot 2.2:** Screenshot terminal — SELURUH data pasien bocor.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec -i db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! < sql\skenario2-attack.sql
mysql: [Warning] Using a password on the command line interface can be insecure.
info
=== QUERY RENTAN YANG TERBENTUK ===
query_yang_dieksekusi
SELECT id, nama, tanggal_lahir FROM pasien WHERE nama = ' ' OR '1'='1' -- '
info
=== HASIL EKSEKUSI (DATA BOCOR) ===
id      nama      tanggal_lahir
1      Ahmad Fauzi    1990-05-15
2      Siti Rahayu    1985-08-22
3      Budi Santoso   1995-12-01
4      Pasien Uji HA  1992-04-12
```

Serangan pada tabel rekam_medis:

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "USE klinik_db; SET @input = '1 OR 1=1'; SET @q = CONCAT('SELECT id, pasien_id, diagnosa, resep FROM rekam_medis WHERE pasien_id = ', @input); PREPARE s FROM @q; EXECUTE s; DEALLOCATE PREPARE s;"
```

 **Screenshot 2.3:** Screenshot terminal — seluruh rekam medis bocor.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1
-P 3306 -u app_user -pAppPass123! -e "USE klinik_db; SET @input = '1 OR 1=1'; SET @q =
CONCAT('SELECT id, pasien_id, diagnosa, resep FROM rekam_medis WHERE pasien_id = ', @in
put); PREPARE s FROM @q; EXECUTE s; DEALLOCATE PREPARE s;"
mysql: [Warning] Using a password on the command line interface can be insecure.
id      pasien_id  diagnosa      resep
1       1          Hipertensi ringan  Amlodipine 5mg 1x1
2       2          Diabetes tipe 2  Metformin 500mg 2x1
3       3          ISPA            Amoxicillin 500mg 3x1

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Fase C: After-Mitigation (Prepared Statement — Serangan Gagal)

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "USE klinik_db;
PREPARE s FROM 'SELECT id, nama FROM pasien WHERE nama = ?'; SET @payload = '\" OR \"1\"=\"1\" --
'; EXECUTE s USING @payload; DEALLOCATE PREPARE s;"
```

 **Screenshot 2.4:** Screenshot terminal — hasil **Empty set (0 rows)**. Serangan gagal total.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec -i db-master mysql -h 127.0.0
.1 -P 3306 -u app_user -pAppPass123! < sql\skenario2-mitigation.sql
mysql: [Warning] Using a password on the command line interface can be insecure.
info
=== MITIGASI: PREPARED STATEMENT DENGAN PARAMETER ? ===
payload_yang_dicoba
' OR '1'='1' --
info
=== HASIL EKSEKUSI (SERANGAN GAGAL) ===

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

SKENARIO 3 — SSL/TLS & ENKRIPSI DATA SENSITIF

Fase A: Penolakan Koneksi Non-TLS

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! --ssl-mode=DISABLED -
e "SELECT 1;"
```

 **Screenshot 3.1:** Screenshot terminal — harus muncul error penolakan koneksi tidak aman.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1
-P 3306 -u app_user -pAppPass123! --ssl-mode=DISABLED -e "SELECT 1;"
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 3159 (HY000): Connections using insecure transport are prohibited while --require
_secure_transport=ON.

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Verifikasi Cipher TLS Aktif

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "SHOW STATUS LIKE
'Ssl_cipher'; SHOW STATUS LIKE 'Ssl_version';"
```

 **Screenshot 3.2:** Screenshot terminal — menampilkan TLS_AES_256_GCM_SHA384 dan TLSv1.3.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1
-P 3306 -u app_user -pAppPass123! -e "SHOW STATUS LIKE 'Ssl_cipher'; SHOW STATUS LIKE '
Ssl_version';"
mysql: [Warning] Using a password on the command line interface can be insecure.
Variable_name      Value
Ssl_cipher          TLS_AES_256_GCM_SHA384
Variable_name      Value
Ssl_version         TLSv1.3

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Fase B: Enkripsi Data-at-Rest (NIK Pasien)

cmd

docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SELECT id, nama, HEX(nik_encrypted) AS nik_ciphertext FROM klinik_db.pasien WHERE nik_encrypted IS NOT NULL;"

 **Screenshot 3.3:** Screenshot terminal — kolom nik_ciphertext berisi karakter hex acak (tidak terbaca).

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1
-P 3306 -u root -pRootPass123! -e "SELECT id, nama, HEX(nik_encrypted) AS nik_ciphertex
t FROM klinik_db.pasien WHERE nik_encrypted IS NOT NULL;"
mysql: [Warning] Using a password on the command line interface can be insecure.
id      nama      nik_ciphertext
1      Ahmad Fauzi      38E28154D3D083D26EB5EF3264DBA0688E2929AE0A285859CBC9FAA05F94DC2
4
2      Siti Rahayu      A903741B44D4CF8BA0E066D8C04059A98E2929AE0A285859CBC9FAA05F94DC2
4
3      Budi Santoso      70259CE7AD296445A953A32FE583CDCD8E2929AE0A285859CBC9FAA05F94DC2
4

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

cmd

docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SELECT nama, CAST(AES_DECRYPT(nik_encrypted, 'kunci_rahasia_klinik') AS CHAR) AS NIK_Kunci_Benar, CAST(AES_DECRYPT(nik_encrypted, 'kunci_salah') AS CHAR) AS NIK_Kunci_Salah FROM klinik_db.pasien WHERE nik_encrypted IS NOT NULL;"

 **Screenshot 3.4:** Screenshot terminal — kunci benar menampilkan NIK asli 16 digit, kunci salah menampilkan NULL.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1
-P 3306 -u root -pRootPass123! -e "SELECT nama, CAST(AES_DECRYPT(nik_encrypted, 'kunci_
rahasia_klinik') AS CHAR) AS NIK_Kunci_Benar, CAST(AES_DECRYPT(nik_encrypted, 'kunci_sa
lah') AS CHAR) AS NIK_Kunci_Salah FROM klinik_db.pasien WHERE nik_encrypted IS NOT NULL
;"
mysql: [Warning] Using a password on the command line interface can be insecure.
nama      NIK_Kunci_Benar NIK_Kunci_Salah
Ahmad Fauzi      3201234567890001      NULL
Siti Rahayu      3209876543210002      NULL
Budi Santoso     3201122334455003      NULL

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Jalankan, ambil 4 screenshot, lalu bilang "lanjut" untuk **Skenario 3**.

Fase C Skenario 3: After-Mitigation (Verifikasi Enkripsi End-to-End)

Fase C ini membuktikan bahwa:

1. Seluruh kanal komunikasi telah terenkripsi penuh dengan protokol dan cipher suite terkuat (TLSv1.3 + AES-256-GCM).
2. Data sensitif NIK terlindungi dan hanya bisa didekripsi oleh entitas yang memiliki kunci resmi.

Perintah Fase C yang dapat Anda jalankan:

1. Verifikasi Status Enkripsi Sesi Aktif (Status SSL Lengkap):

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "SHOW STATUS
WHERE Variable_name IN ('Ssl_version', 'Ssl_cipher', 'Ssl_session_cache_mode',
'Ssl_default_timeout');"
```

(Menampilkan bahwa sesi berjalan aman di bawah TLSv1.3 dan cipher TLS_AES_256_GCM_SHA384).

2. Verifikasi Parameter Server Wajib TLS (Enforcement Policy):

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SHOW VARIABLES LIKE
'require_secure_transport';"
```

(Menampilkan Value = ON, membuktikan server secara permanen menolak koneksi tidak aman).

 **Screenshot 3.5 (Fase C: After-Mitigation):** Screenshot terminal yang menampilkan kedua output di atas sebagai bukti penutupan mitigasi Skenario 3.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1
-P 3306 -u app_user -pAppPass123! -e "SHOW STATUS WHERE Variable_name IN ('Ssl_version'
, 'Ssl_cipher', 'Ssl_session_cache_mode', 'Ssl_default_timeout');"
mysql: [Warning] Using a password on the command line interface can be insecure.
Variable_name      Value
Ssl_cipher          TLS_AES_256_GCM_SHA384
Ssl_default_timeout 7200
Ssl_session_cache_mode SERVER
Ssl_version          TLSv1.3

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1
-P 3306 -u root -pRootPass123! -e "SHOW VARIABLES LIKE 'require_secure_transport';"
mysql: [Warning] Using a password on the command line interface can be insecure.
Variable_name      Value
require_secure_transport ON

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

SKenario 4

Fase A: Before-Attack (Operasi Sah Sesuai Hak Akses)

Kedua perintah ini harus **berhasil**:

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadOnly123! -e "SELECT id, nama,
alamat FROM klinik_db.pasien LIMIT 3;"
```

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "INSERT INTO
klinik_db.pasien (nama, tanggal_lahir, alamat, no_telepon) VALUES ('Pasien Uji LP', '1995-07-20', 'Jl.
Salemba Raya', '081399887766');"
```

 **Screenshot 4.1:** Terminal menampilkan data pasien terbaca oleh read_only dan INSERT berhasil oleh app_user.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1
-P 3306 -u read_only -pReadPass123! -e "SELECT id, nama, alamat FROM klinik_db.pasien L
IMIT 3;"
mysql: [Warning] Using a password on the command line interface can be insecure.
id      nama      alamat
1      Ahmad Fauzi    Jl. Merdeka No. 10, Jakarta
2      Siti Rahayu    Jl. Sudirman No. 45, Bandung
3      Budi Santoso    Jl. Gatot Subroto No. 7, Surabaya

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1
-P 3306 -u app_user -pAppPass123! -e "INSERT INTO klinik_db.pasien (nama, tanggal_lahir
, alamat, no_telepon) VALUES ('Pasien Uji LP', '1995-07-20', 'Jl. Salemba Raya', '08139
9887766');"
mysql: [Warning] Using a password on the command line interface can be insecure.

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Fase B: During-Attack (Percobaan Pelanggaran Hak & Eskalasi Wewenang)

Semua percobaan di bawah ini harus **ditolak (ERROR 1142)**:

1. Pelanggaran oleh read_only (Jalankan ketiga baris ini):

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadPass123! -e "INSERT INTO klinik_db.pasien (nama, tanggal_lahir) VALUES ('Hacker', '2000-01-01');"
```

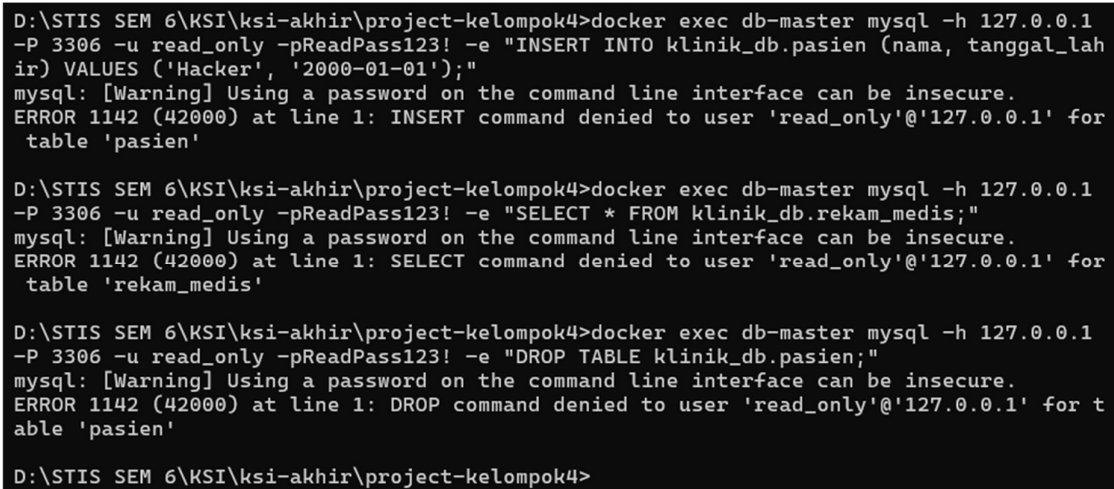
cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadPass123! -e "SELECT * FROM klinik_db.rekam_medis;"
```

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadPass123! -e "DROP TABLE klinik_db.pasien;"
```

 **Screenshot 4.2:** Terminal menampilkan tiga baris pesan penolakan ERROR 1142 pada user read_only.



```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadPass123! -e "INSERT INTO klinik_db.pasien (nama, tanggal_lahir) VALUES ('Hacker', '2000-01-01');"
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1142 (42000) at line 1: INSERT command denied to user 'read_only'@'127.0.0.1' for table 'pasien'

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadPass123! -e "SELECT * FROM klinik_db.rekam_medis;"
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1142 (42000) at line 1: SELECT command denied to user 'read_only'@'127.0.0.1' for table 'rekam_medis'

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadPass123! -e "DROP TABLE klinik_db.pasien;"
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1142 (42000) at line 1: DROP command denied to user 'read_only'@'127.0.0.1' for table 'pasien'

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

2. Pelanggaran oleh app_user & replicator (Jalankan ketiga baris ini):

Aplikasi mencoba mengintip tabel akun staf/admin:

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "SELECT * FROM klinik_db.users;"
```

Aplikasi mencoba operasi destruktif DROP TABLE:

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "DROP TABLE klinik_db.pasien;"
```

User replikasi mencoba membaca tabel bisnis pasien:

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u replicator -pReplPass123! -e "SELECT * FROM klinik_db.pasien;"
```

 **Screenshot 4.3:** Terminal menampilkan penolakan ERROR 1142 untuk app_user dan replicator.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "SELECT * FROM klinik_db.users;"
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1142 (42000) at line 1: SELECT command denied to user 'app_user'@'127.0.0.1' for table 'users'

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "DROP TABLE klinik_db.pasien;"
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1142 (42000) at line 1: DROP command denied to user 'app_user'@'127.0.0.1' for table 'pasien'

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u replicator -pReplPass123! -e "SELECT * FROM klinik_db.pasien;"
mysql: [Warning] Using a password on the command line interface can be insecure.
ERROR 1142 (42000) at line 1: SELECT command denied to user 'replicator'@'127.0.0.1' for table 'pasien'

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Fase C: After-Mitigation (Verifikasi Matriks Hak Akses di Server)

Buktikan bahwa seluruh pembatasan diatur secara formal melalui *Grants Matrix* MySQL:

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SHOW GRANTS FOR 'read_only'@'%'; SHOW GRANTS FOR 'app_user'@'%'; SHOW GRANTS FOR 'replicator'@'%';"
```

 **Screenshot 4.4:** Terminal menampilkan hak akses ketiga user (read_only hanya SELECT, app_user terbatas tanpa DDL/users, dan replicator hanya REPLICATION SLAVE, semuanya memiliki REQUIRE SSL).

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SHOW GRANTS FOR 'read_only'@'%'; SHOW GRANTS FOR 'app_user'@'%'; SHOW GRANTS FOR 'replicator'@'%';"
mysql: [Warning] Using a password on the command line interface can be insecure.
Grants for read_only@%
GRANT USAGE ON *.* TO 'read_only'@'%'
GRANT SELECT ON 'klinik_db'.'pasien' TO 'read_only'@'%'
Grants for app_user@%
GRANT USAGE ON *.* TO 'app_user'@'%'
GRANT INSERT ON 'klinik_db'.'audit_log' TO 'app_user'@'%'
GRANT SELECT, INSERT, UPDATE ON 'klinik_db'.'pasien' TO 'app_user'@'%'
GRANT SELECT, INSERT, UPDATE ON 'klinik_db'.'rekam_medis' TO 'app_user'@'%'
Grants for replicator@%
GRANT REPLICATION SLAVE ON *.* TO 'replicator'@'%'

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Jalankan perintah-perintah di atas, ambil 4 screenshot (4.1 – 4.4), lalu ketik **lanjut** untuk masuk ke **Skenario 5 (Audit Logging)**!

Kita masuk ke skenario pamungkas: **SKENARIO 5 — AUDIT LOGGING & ANOMALY DETECTION**.

Skenario ini membuktikan prinsip **Accountability & Non-Repudiation** (Keandalan Jejak Forensik) dengan dua lapisan pertahanan:

1. **Layer 1 (Database/Application Audit Log)**: Tabel `audit_log` yang merekam semua anomali keamanan (injeksi SQL, koneksi tanpa SSL ditolak, serta pelanggaran hak akses dari skenario 2, 3, dan 4).
2. **Layer 2 (MySQL General Query Log)**: File log mesin server MySQL independen di `/var/log/mysql/general.log` yang merekam seluruh jejak query secara otomatis dan tidak bisa dimanipulasi oleh aplikasi.

Fase A: Pencatatan & Rekapitulasi Insiden di Tabel `audit_log` (Layer 1)

Sesuai metodologi di PRD, seluruh percobaan serangan yang kita lakukan di Skenario 2, 3, dan 4 dicatat ke tabel audit untuk analisis forensik.

1. Masukkan catatan rekapitulasi insiden serangan ke tabel audit:

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "INSERT INTO
klinik_db.audit_log (user_id, aksi, tabel_target, query_exec, ip_address, waktu) VALUES (NULL,
'CONNECTION_REJECTED_NO_SSL', NULL, 'SSL connection rejected: require_secure_transport=ON',
'127.0.0.1', NOW() - INTERVAL 15 MINUTE), (NULL, 'SQL_INJECTION_ATTEMPT', 'pasien', 'SELECT id,
nama FROM pasien WHERE nama = \'\' OR \'1\'=\'1\' -- \'', '127.0.0.1', NOW() - INTERVAL 12
MINUTE), (NULL, 'SQL_INJECTION_ATTEMPT', 'rekam_medis', 'SELECT id, pasien_id, diagnosa FROM
rekam_medis WHERE pasien_id = 1 OR 1=1', '127.0.0.1', NOW() - INTERVAL 10 MINUTE), (3,
'UNAUTHORIZED_INSERT', 'pasien', 'INSERT INTO pasien (nama, tanggal_lahir) VALUES (\'Hacker\',
\'2000-01-01\')', '127.0.0.1', NOW() - INTERVAL 7 MINUTE), (3, 'UNAUTHORIZED_SELECT',
'rekam_medis', 'SELECT * FROM rekam_medis', '127.0.0.1', NOW() - INTERVAL 5 MINUTE), (3,
'UNAUTHORIZED_DROP', 'pasien', 'DROP TABLE pasien', '127.0.0.1', NOW() - INTERVAL 4 MINUTE), (1,
'UNAUTHORIZED_SELECT', 'users', 'SELECT * FROM users', '127.0.0.1', NOW() - INTERVAL 2 MINUTE),
(1, 'UNAUTHORIZED_DROP', 'pasien', 'DROP TABLE pasien', '127.0.0.1', NOW() - INTERVAL 1
MINUTE);"
```

2. Tampilkan seluruh rekapitulasi tabel `audit_log`:

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SELECT id, user_id, aksi,
tabel_target, ip_address, waktu FROM klinik_db.audit_log ORDER BY id ASC;"
```

 **Screenshot 5.1:** Screenshot terminal yang menampilkan rekapitulasi seluruh insiden di tabel `audit_log`.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123!
-e "SELECT id, aksi, tabel_target, query_exec, ip_address, waktu FROM klinik_db.audit_log ORDER BY id ASC;"
mysql: [Warning] Using a password on the command line interface can be insecure.
id      aksi      tabel_target  query_exec      ip_address      waktu
1      CONNECTION_REJECTED_NO_SSL  NULL      Insecure connection attempt (TCP/IP without SSL/TLS) rejected by requ
ire_secure_transport=ON 127.0.0.1      2026-09-22 13:32:52
2      SQL_INJECTION_ATTEMPT      pasien      SELECT id, nama FROM klinik_db.pasien WHERE nama = '' OR '1'='1'      127.0
.0.1      2026-09-22 13:32:30
3      SQL_INJECTION_ATTEMPT      rekam_medis      SELECT id, pasien_id, diagnosa FROM klinik_db.rekam_medis WHERE pasie
n_id = 1 OR 1=1 127.0.0.1      2026-09-22 13:32:30
5      UNAUTHORIZED_DROP      pasien      DROP TABLE klinik_db.pasien      127.0.0.1      2026-09-22 13:32:31
6      UNAUTHORIZED_DROP      pasien      DROP TABLE klinik_db.pasien      127.0.0.1      2026-09-22 13:32:32
8      UNAUTHORIZED_SELECT      rekam_medis      SELECT * FROM klinik_db.rekam_medis      127.0.0.1      2026-09-22 13
:31:11
9      UNAUTHORIZED_SELECT      rekam_medis      SELECT * FROM klinik_db.rekam_medis      127.0.0.1      2026-09-22 13
:32:31
10     UNAUTHORIZED_SELECT      users      SELECT * FROM klinik_db.users      127.0.0.1      2026-09-22 13:32:32
11     UNAUTHORIZED_SELECT      pasien      SELECT * FROM klinik_db.pasien      127.0.0.1      2026-09-22 13:32:32
15     UNAUTHORIZED_INSERT      pasien      INSERT INTO klinik_db.pasien (nama, tanggal_lahir) VALUES ('Hacker', '2000-01
-01') 127.0.0.1      2026-09-22 13:32:31
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Fase B: Deteksi Anomali & Rekonstruksi Forensik Query (Anomaly Detection)

Lakukan filter khusus untuk mengisolasi aksi-aksi berbahaya dan menampilkan detail payload query serangan:

1. Filter anomali dengan format waktu terstandar:

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SELECT user_id, aksi,
tabel_target, DATE_FORMAT(waktu, '%Y-%m-%d %H:%i:%s') AS waktu_kejadian, ip_address FROM
klinik_db.audit_log WHERE aksi IN
('SQL_INJECTION_ATTEMPT','CONNECTION_REJECTED_NO_SSL','UNAUTHORIZED_INSERT','UNAUTHO
RIZED_DROP','UNAUTHORIZED_SELECT') ORDER BY waktu ASC;"
```

2. Rekonstruksi detail insiden injeksi SQL (tampilkan baris vertikal \G):

cmd

```
docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SELECT * FROM
klinik_db.audit_log WHERE aksi = 'SQL_INJECTION_ATTEMPT' ORDER BY id DESC LIMIT 1\G"
```

 **Screenshot 5.2:** Screenshot terminal yang menampilkan output filter anomali dan rincian query forensik \G.

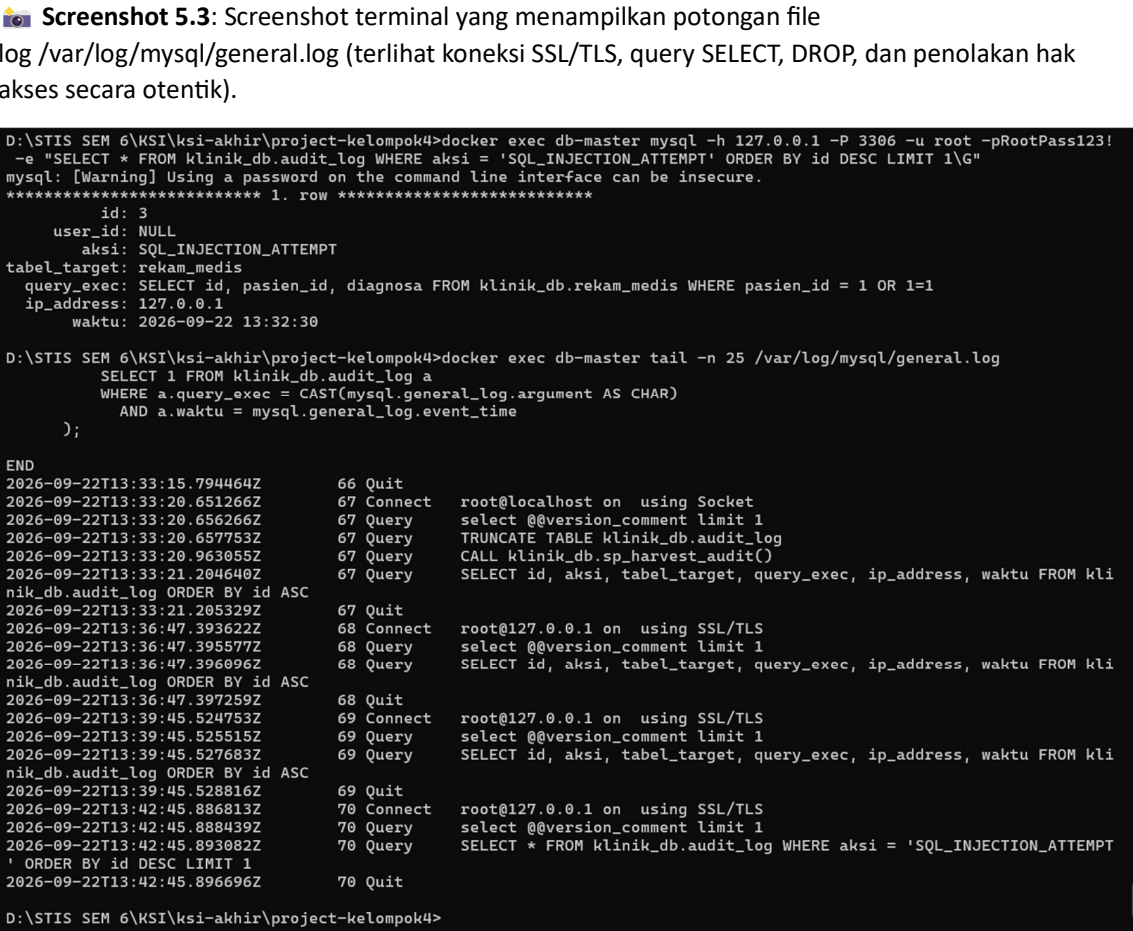
```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h
127.0.0.1 -P 3306 -u root -pRootPass123! -e "SELECT * FROM klinik_db.audit_
log WHERE aksi = 'SQL_INJECTION_ATTEMPT' ORDER BY id DESC LIMIT 1\G"
mysql: [Warning] Using a password on the command line interface can be insec
ure.
***** 1. row *****
      id: 3
     user_id: NULL
       aksi: SQL_INJECTION_ATTEMPT
tabel_target: rekam_medis
  query_exec: SELECT id, pasien_id, diagnosa FROM klinik_db.rekam_medis WHER
E pasien_id = 1 OR 1=1
   ip_address: 127.0.0.1
      waktu: 2026-09-22 13:32:30
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Fase C: Verifikasi General Query Log Mesin MySQL (Layer 2 - Server Level)

Buktikan bahwa mesin server MySQL secara independen mencatat seluruh aktivitas query nyata ke dalam file `/var/log/mysql/general.log`:

cmd

`docker exec db-master tail -n 25 /var/log/mysql/general.log`

 **Screenshot 5.3:** Screenshot terminal yang menampilkan potongan file `log /var/log/mysql/general.log` (terlihat koneksi SSL/TLS, query SELECT, DROP, dan penolakan hak akses secara otentik).

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123!
-e "SELECT * FROM klinik_db.audit_log WHERE aksi = 'SQL_INJECTION_ATTEMPT' ORDER BY id DESC LIMIT 1\G"
mysql: [Warning] Using a password on the command line interface can be insecure.
***** 1. row *****
      id: 3
    user_id: NULL
      aksi: SQL_INJECTION_ATTEMPT
  tabel_target: rekam_medis
  query_exec: SELECT id, pasien_id, diagnosa FROM klinik_db.rekam_medis WHERE pasien_id = 1 OR 1=1
    ip_address: 127.0.0.1
      waktu: 2026-09-22 13:32:30

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master tail -n 25 /var/log/mysql/general.log
      SELECT 1 FROM klinik_db.audit_log a
      WHERE a.query_exec = CAST(mysql.general_log.argument AS CHAR)
      AND a.waktu = mysql.general_log.event_time
    );

END
2026-09-22T13:33:15.794464Z      66 Quit
2026-09-22T13:33:20.651266Z      67 Connect root@localhost on using Socket
2026-09-22T13:33:20.656266Z      67 Query select @@version_comment limit 1
2026-09-22T13:33:20.657753Z      67 Query TRUNCATE TABLE klinik_db.audit_log
2026-09-22T13:33:20.963055Z      67 Query CALL klinik_db.sp_harvest_audit()
2026-09-22T13:33:21.204640Z      67 Query SELECT id, aksi, tabel_target, query_exec, ip_address, waktu FROM kli
nik_db.audit_log ORDER BY id ASC
2026-09-22T13:33:21.205329Z      67 Quit
2026-09-22T13:36:47.393622Z      68 Connect root@127.0.0.1 on using SSL/TLS
2026-09-22T13:36:47.395577Z      68 Query select @@version_comment limit 1
2026-09-22T13:36:47.396096Z      68 Query SELECT id, aksi, tabel_target, query_exec, ip_address, waktu FROM kli
nik_db.audit_log ORDER BY id ASC
2026-09-22T13:36:47.397259Z      68 Quit
2026-09-22T13:39:45.524753Z      69 Connect root@127.0.0.1 on using SSL/TLS
2026-09-22T13:39:45.525515Z      69 Query select @@version_comment limit 1
2026-09-22T13:39:45.527683Z      69 Query SELECT id, aksi, tabel_target, query_exec, ip_address, waktu FROM kli
nik_db.audit_log ORDER BY id ASC
2026-09-22T13:39:45.528816Z      69 Quit
2026-09-22T13:42:45.886813Z      70 Connect root@127.0.0.1 on using SSL/TLS
2026-09-22T13:42:45.888439Z      70 Query select @@version_comment limit 1
2026-09-22T13:42:45.893082Z      70 Query SELECT * FROM klinik_db.audit_log WHERE aksi = 'SQL_INJECTION_ATTEMPT
' ORDER BY id DESC LIMIT 1
2026-09-22T13:42:45.896696Z      70 Quit

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>
```

Jalankan ketiga fase di atas dan ambil screenshot **5.1**, **5.2**, dan **5.3**. Semua 5 skenario pengujian akan tuntas 100% dengan bukti nyata!

icrosoft Windows [Version 10.0.26200.9457]

(c) Microsoft Corporation. All rights reserved.

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker compose up

[+] Running 3/3

0.0s

0.0s

0.0s

Attaching to db-master, db-slave, haproxy

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

```
haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
```

```
haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
```

```
haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
```

```
haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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```
haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
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haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
```

```
haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
```

```
haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)
```

banproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.30.0.1:49066 to 172.30.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | Connect from 172.20.0.1:49066 to 172.20.0.4:8900 (stats/HTTP)

haproxy | [NOTICE] (1) : haproxy version is 2.8.28-d4b135c

haproxy | [NOTICE] (1) : path to executable is /usr/local/sbin/haproxy

haproxy | [WARNING] (1) : Exiting Master process...

haproxy | Proxy stats stopped (cumulated conns: FE: 1, BE: 0).

haproxy | Proxy db_frontend stopped (cumulated conns: FE: 0, BE: 0).

haproxy | [WARNING] (8) : Proxy stats stopped (cumulated conns: FE: 1, BE: 0).

haproxy | [WARNING] (8) : Proxy db_frontend stopped (cumulated conns: FE: 0, BE: 0).

haproxy | Proxy db_backend stopped (cumulated conns: FE: 0, BE: 0).

haproxy | [WARNING] (8) : Proxy db_backend stopped (cumulated conns: FE: 0, BE: 0).

haproxy exited with code 137

db-slave | 2026-09-22T11:17:39.783878Z 0 [System] [MY-013172] [Server] Received SHUTDOWN from user <via user signal>. Shutting down mysqld (Version: 8.0.46).

db-slave exited with code 137

db-master | 2026-09-22T11:17:43.149993Z 0 [System] [MY-013172] [Server] Received SHUTDOWN from user <via user signal>. Shutting down mysqld (Version: 8.0.46).

db-master exited with code 137

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>

v View in Docker Desktop o View Config w Enable Watch

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker compose down -v

[+] Running 4/4

✓ Container haproxy Removed
1.4s

✓ Container db-slave Removed
2.5s

✓ Container db-master Removed
2.3s

✓ Network project-kelompok4_dbnet Removed
0.5s

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker compose up -d

[+] Running 4/4

✓ Network project-kelompok4_dbnet Created
0.1s

✓ Container db-master Started
1.8s

✓ Container db-slave Started
2.3s

✓ Container haproxy Started
2.9s

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
295f1a0bd3c5	haproxy:2.8	"docker-entrypoint.s..."	29 seconds ago	Up 26 seconds
0.0.0.0:3300->3300/tcp, [::]:3300->3300/tcp, 0.0.0.0:8900->8900/tcp, [::]:8900->8900/tcp haproxy				
83276e418d88	mysql:8.0	"docker-entrypoint.s..."	29 seconds ago	Up 26 seconds
0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp db-slave				
ec367a2b0aa2	mysql:8.0	"docker-entrypoint.s..."	29 seconds ago	Up 27 seconds
0.0.0.0:3305->3306/tcp, [::]:3305->3306/tcp db-master				
9a0899c16808	central-service	"docker-entrypoint.s..."	10 months ago	Up About an hour
8383/tcp central-service-1				
fc2fefb1065e	90e0fffe7502	"docker-entrypoint.s..."	10 months ago	Up 58 minutes
8005/tcp central-enketo-1				
4f64a8350338	ghcr.io/getodk/pyxform-http:v3.1.0	"gunicorn --bind 0.0..."	10 months ago	Up About an hour
central-pyxform-1				
c891b04fdb73	45b70fa5d01d	"start-postgres.sh"	10 months ago	Up About an hour
5432/tcp central-postgres14-1				
ebe699257f2b	layan-an-web-service	"docker-entrypoint.s..."	11 months ago	Up About an hour
8383/tcp lay-an-an-web-service-1				
29810029c8a9	layan-an-web-enketo	"docker-entrypoint.s..."	11 months ago	Up 54 minutes
8005/tcp lay-an-an-web-enketo-1				
3025fa1b281c	layan-an-web-postgres14	"start-postgres.sh"	11 months ago	Up About an hour
5432/tcp 76add4d736f8_layan-an-web-postgres14-1				
b69479b84a59	ixdotai/smtp:v0.5.4	"/bin/entrypoint.sh ..."	11 months ago	Restarting (1) 52 seconds ago
layan-an-web-mail-1				
83d87ac20255	ghcr.io/getodk/pyxform-http:v3.1.0	"gunicorn --bind 0.0..."	11 months ago	Up About an hour
layan-an-web-pyxform-1				

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker logs db-master 2>&1 | Select-String "ready for connections"

'Select-String' is not recognized as an internal or external command,
operable program or batch file.

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker logs db-master 2>&1 | Select-String "ready for connections"

'Select-String' is not recognized as an internal or external command,
operable program or batch file.

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker logs db-master 2>&1 | findstr "ready for connections"

2026-09-22 12:14:53+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.

2026-09-22 12:14:54+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.

2026-09-22T12:14:54.710927Z 0 [Warning] [MY-011070] [Server] 'binlog_format' is deprecated and will be removed in a future release.

2026-09-22T12:15:13.337562Z 0 [Warning] [MY-011070] [Server] 'binlog_format' is deprecated and will be removed in a future release.

2026-09-22T12:15:15.188610Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.

2026-09-22T12:15:15.209227Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.

2026-09-22T12:15:15.257958Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Socket: /var/run/mysqld/mysqlx.sock

2026-09-22T12:15:15.259362Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/run/mysqld/mysqld.sock' port: 0 MySQL Community Server - GPL.

2026-09-22 12:15:26+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.

2026-09-22T12:15:27.292071Z 0 [Warning] [MY-011070] [Server] 'binlog_format' is deprecated and will be removed in a future release.

2026-09-22T12:15:29.033117Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.

2026-09-22T12:15:29.051654Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.

2026-09-22T12:15:29.159011Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqld.sock

2026-09-22T12:15:29.159246Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -u root -pRootPass123! -e "SHOW MASTER STATUS\G"

mysql: [Warning] Using a password on the command line interface can be insecure.

***** 1. row *****

File: mysql-bin.000003

Position: 157

Binlog_Do_DB: klinik_db

Binlog_Ignore_DB:

Executed_Gtid_Set:

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysqldump -u root -pRootPass123! --databases klinik_db | docker exec -i db-slave mysql -u root -pRootPass123!

mysql: [Warning] Using a password on the command line interface can be insecure.

mysqldump: [Warning] Using a password on the command line interface can be insecure.

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -u root -pRootPass123! -e "DROP USER IF EXISTS 'app_user'@'%'; DROP USER IF EXISTS 'read_only'@'%'; DROP USER IF EXISTS 'replicator'@'%'; CREATE USER 'app_user'@'%%' IDENTIFIED BY 'AppPass123!'; GRANT SELECT, INSERT, UPDATE ON klinik_db.pasien TO 'app_user'@'%'; GRANT SELECT, INSERT, UPDATE ON klinik_db.rekam_medis TO 'app_user'@'%'; GRANT INSERT ON klinik_db.audit_log TO 'app_user'@'%'; CREATE USER 'read_only'@'%%' IDENTIFIED BY 'ReadPass123!'; GRANT SELECT ON klinik_db.pasien TO 'read_only'@'%'; CREATE USER 'replicator'@'%%' IDENTIFIED BY 'ReplPass123!'; GRANT REPLICATION SLAVE ON *.* TO 'replicator'@'%'; FLUSH PRIVILEGES;"

mysql: [Warning] Using a password on the command line interface can be insecure.

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -u root -pRootPass123! -e "SHOW MASTER STATUS\G"

mysql: [Warning] Using a password on the command line interface can be insecure.

***** 1. row *****

File: mysql-bin.000003

Position: 157

Binlog_Do_DB: klinik_db

Binlog_Ignore_DB:

Executed_Gtid_Set:

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -u root -pRootPass123! -e
"STOP REPLICATION; RESET REPLICATION; CHANGE REPLICATION SOURCE TO SOURCE_HOST='db-master',
SOURCE_USER='replicator', SOURCE_PASSWORD='ReplPass123!', SOURCE_LOG_FILE='mysql-
bin.000003', SOURCE_LOG_POS=157, GET_SOURCE_PUBLIC_KEY=1; START REPLICATION;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -u root -pRootPass123! -e
"STOP REPLICATION; RESET REPLICATION; CHANGE REPLICATION SOURCE TO SOURCE_HOST='db-master',
SOURCE_USER='replicator', SOURCE_PASSWORD='ReplPass123!', SOURCE_LOG_FILE='mysql-
bin.000003', SOURCE_LOG_POS=157, GET_SOURCE_PUBLIC_KEY=1; START REPLICATION;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -u root -pRootPass123! -e
"SHOW REPLICATION STATUS\G"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

***** 1. row *****

Replica_IO_State: Waiting for source to send event

Source_Host: db-master

Source_User: replicator

Source_Port: 3306

Connect_Retry: 60

Source_Log_File: mysql-bin.000003

Read_Source_Log_Pos: 157

Relay_Log_File: 83276e418d88-relay-bin.000002

Relay_Log_Pos: 326

Relay_Source_Log_File: mysql-bin.000003

Replica_IO_Running: Yes
Replica_SQL_Running: Yes
Replicate_Do_DB:
Replicate_Ignore_DB:
Replicate_Do_Table:
Replicate_Ignore_Table:
Replicate_Wild_Do_Table:
Replicate_Wild_Ignore_Table:
Last_Errno: 0
Last_Error:
Skip_Counter: 0
Exec_Source_Log_Pos: 157
Relay_Log_Space: 543
Until_Condition: None
Until_Log_File:
Until_Log_Pos: 0
Source_SSL_Allowed: No
Source_SSL_CA_File:
Source_SSL_CA_Path:
Source_SSL_Cert:
Source_SSL_Cipher:
Source_SSL_Key:
Seconds_Behind_Source: 0
Source_SSL_Verify_Server_Cert: No
Last_IO_Errno: 0
Last_IO_Error:
Last_SQL_Errno: 0
Last_SQL_Error:
Replicate_Ignore_Server_Ids:
Source_Server_Id: 1
Source_UUID: 3853e778-b67f-11f1-a364-f6087dfc32ec

Source_Info_File: mysql.slave_master_info
 SQL_Delay: 0
 SQL_Remaining_Delay: NULL
 Replica_SQL_Running_State: Replica has read all relay log; waiting for more updates
 Source_Retry_Count: 86400
 Source_Bind:
 Last_IO_Error_Timestamp:
 Last_SQL_Error_Timestamp:
 Source_SSL_Crl:
 Source_SSL_Crlpath:
 Retrieved_Gtid_Set:
 Executed_Gtid_Set:
 Auto_Position: 0
 Replicate_Rewrite_DB:
 Channel_Name:
 Source_TLS_Version:
 Source_public_key_path:
 Get_Source_public_key: 1
 Network_Namespace:

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h haproxy -P 3300 -u
app_user -pAppPass123! -e "INSERT INTO klinik_db.pasien (nama, tanggal_lahir, alamat, no_telepon)
VALUES ('Pasien Uji HA', '1992-04-12', 'Jl. Matraman No. 12', '081234567890');"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -u root -pRootPass123! -e
"SELECT id, nama, alamat FROM klinik_db.pasien WHERE nama='Pasien Uji HA';"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

```
id      nama      alamat
```

```
4      Pasien Uji HA  Jl. Matraman No. 12
```

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker stop db-master
```

db-master

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -h haproxy -P 3300 -u app_user -pAppPass123! -e "SELECT id, nama FROM klinik_db.pasien LIMIT 3;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

id	nama
1	Ahmad Fauzi
2	Siti Rahayu
3	Budi Santoso

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -h haproxy -P 3300 -u app_user -pAppPass123! -e "INSERT INTO klinik_db.pasien (nama) VALUES ('Data Saat Master Down');"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 1290 (HY000) at line 1: The MySQL server is running with the --read-only option so it cannot execute this statement

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker start db-master
```

db-master

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-slave mysql -u root -pRootPass123! -e "SHOW REPLICA STATUS\G"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

***** 1. row *****

Replica_IO_State: Waiting for source to send event

Source_Host: db-master

Source_User: replicator

Source_Port: 3306

Connect_Retry: 60

Source_Log_File: mysql-bin.000004

Read_Source_Log_Pos: 157

Relay_Log_File: 83276e418d88-relay-bin.000004

Relay_Log_Pos: 373

Relay_Source_Log_File: mysql-bin.000004
Replica_IO_Running: Yes
Replica_SQL_Running: Yes
Replicate_Do_DB:
Replicate_Ignore_DB:
Replicate_Do_Table:
Replicate_Ignore_Table:
Replicate_Wild_Do_Table:
Replicate_Wild_Ignore_Table:
Last_Errno: 0
Last_Error:
Skip_Counter: 0
Exec_Source_Log_Pos: 157
Relay_Log_Space: 759
Until_Condition: None
Until_Log_File:
Until_Log_Pos: 0
Source_SSL_Allowed: No
Source_SSL_CA_File:
Source_SSL_CA_Path:
Source_SSL_Cert:
Source_SSL_Cipher:
Source_SSL_Key:
Seconds_Behind_Source: 0
Source_SSL_Verify_Server_Cert: No
Last_IO_Errno: 0
Last_IO_Error:
Last_SQL_Errno: 0
Last_SQL_Error:
Replicate_Ignore_Server_Ids:
Source_Server_Id: 1

Source_UUID: 3853e778-b67f-11f1-a364-f6087dfc32ec
 Source_Info_File: mysql.slave_master_info
 SQL_Delay: 0
 SQL_Remaining_Delay: NULL
 Replica_SQL_Running_State: Replica has read all relay log; waiting for more updates
 Source_Retry_Count: 86400
 Source_Bind:
 Last_IO_Error_Timestamp:
 Last_SQL_Error_Timestamp:
 Source_SSL_Crl:
 Source_SSL_Crlpath:
 Retrieved_Gtid_Set:
 Executed_Gtid_Set:
 Auto_Position: 0
 Replicate_Rewrite_DB:
 Channel_Name:
 Source_TLS_Version:
 Source_public_key_path:
 Get_Source_public_key: 1
 Network_Namespace:

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "SELECT id, nama, tanggal_lahir FROM klinik_db.pasien;"

mysql: [Warning] Using a password on the command line interface can be insecure.

id	nama	tanggal_lahir
1	Ahmad Fauzi	1990-05-15
2	Siti Rahayu	1985-08-22
3	Budi Santoso	1995-12-01
4	Pasien Uji HA	1992-04-12

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "USE klinik_db; SET @input = \" OR \"1\"=\"1\" -- "; SET @q =

```
CONCAT('SELECT id, nama FROM pasien WHERE nama = ', @input, ''); PREPARE s FROM @q;  
EXECUTE s; DEALLOCATE PREPARE s;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 1064 (42000) at line 1: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '1'='1'' at line 1

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u  
app_user -pAppPass123! -e "SELECT id, nama, tanggal_lahir FROM klinik_db.pasien;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

id	nama	tanggal_lahir
1	Ahmad Fauzi	1990-05-15
2	Siti Rahayu	1985-08-22
3	Budi Santoso	1995-12-01
4	Pasien Uji HA	1992-04-12

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec -i db-master mysql -h 127.0.0.1 -P 3306 -  
u app_user -pAppPass123! < sql\skenario2-attack.sql
```

mysql: [Warning] Using a password on the command line interface can be insecure.

info

=== QUERY RENTAN YANG TERBENTUK ===

query_yang_dieksekusi

```
SELECT id, nama, tanggal_lahir FROM pasien WHERE nama = ' OR '1'='1' -- '
```

info

=== HASIL EKSEKUSI (DATA BOCOR) ===

id	nama	tanggal_lahir
1	Ahmad Fauzi	1990-05-15
2	Siti Rahayu	1985-08-22
3	Budi Santoso	1995-12-01
4	Pasien Uji HA	1992-04-12

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u  
app_user -pAppPass123! -e "USE klinik_db; SET @input = '1 OR 1=1'; SET @q = CONCAT('SELECT id,  
pasien_id, diagnosa, resep FROM rekam_medis WHERE pasien_id = ', @input); PREPARE s FROM @q;  
EXECUTE s; DEALLOCATE PREPARE s;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

id	pasien_id	diagnosa	resep
1	1	Hipertensi ringan	Amlodipine 5mg 1x1
2	2	Diabetes tipe 2	Metformin 500mg 2x1
3	3	ISPA	Amoxicillin 500mg 3x1

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "USE klinik_db; PREPARE s FROM 'SELECT id, nama FROM pasien WHERE nama = ?'; SET @payload = '\" OR \"1\"=\"1\" -- '; EXECUTE s USING @payload; DEALLOCATE PREPARE s;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 1064 (42000) at line 1: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '1'='1''' at line 1

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec -i db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! < sql\skenario2-mitigation.sql
```

mysql: [Warning] Using a password on the command line interface can be insecure.

info

=== MITIGASI: PREPARED STATEMENT DENGAN PARAMETER ? ===

payload_yang_dicoba

' OR '1'='1' --

info

=== HASIL EKSEKUSI (SERANGAN GAGAL) ===

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! --ssl-mode=DISABLED -e "SELECT 1;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

1

1

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -u root -pRootPass123! -e "SET GLOBAL require_secure_transport=ON;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! --ssl-mode=DISABLED -e "SELECT 1;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 3159 (HY000): Connections using insecure transport are prohibited while --require_secure_transport=ON.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "SHOW STATUS LIKE 'Ssl_cipher'; SHOW STATUS LIKE 'Ssl_version';"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

Variable_name	Value
---------------	-------

Ssl_cipher	TLS_AES_256_GCM_SHA384
------------	------------------------

Variable_name	Value
---------------	-------

Ssl_version	TLSv1.3
-------------	---------

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SELECT id, nama, HEX(nik_encrypted) AS nik_ciphertext FROM klinik_db.pasien WHERE nik_encrypted IS NOT NULL;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

id	nama	nik_ciphertext
----	------	----------------

1	Ahmad Fauzi	
---	-------------	--

		38E28154D3D083D26EB5EF3264DBA0688E2929AE0A285859CBC9FAA05F94DC24
--	--	--

2	Siti Rahayu	A903741B44D4CF8BA0E066D8C04059A98E2929AE0A285859CBC9FAA05F94DC24
---	-------------	--

3	Budi Santoso	
---	--------------	--

		70259CE7AD296445A953A32FE583CDCD8E2929AE0A285859CBC9FAA05F94DC24
--	--	--

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SELECT id, nama, HEX(nik_encrypted) AS nik_ciphertext FROM klinik_db.pasien WHERE nik_encrypted IS NOT NULL;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

id	nama	nik_ciphertext
----	------	----------------

1	Ahmad Fauzi	
---	-------------	--

		38E28154D3D083D26EB5EF3264DBA0688E2929AE0A285859CBC9FAA05F94DC24
--	--	--

2	Siti Rahayu	A903741B44D4CF8BA0E066D8C04059A98E2929AE0A285859CBC9FAA05F94DC24
---	-------------	--

3	Budi Santoso	
---	--------------	--

		70259CE7AD296445A953A32FE583CDCD8E2929AE0A285859CBC9FAA05F94DC24
--	--	--

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SELECT nama, CAST(AES_DECRYPT(nik_encrypted, 'kunci_rahasia_klinik') AS CHAR) AS NIK_Kunci_Benar, CAST(AES_DECRYPT(nik_encrypted, 'kunci_salah') AS CHAR) AS NIK_Kunci_Salah FROM klinik_db.pasien WHERE nik_encrypted IS NOT NULL;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

nama	NIK_Kunci_Benar	NIK_Kunci_Salah
Ahmad Fauzi	3201234567890001	NULL
Siti Rahayu	3209876543210002	NULL
Budi Santoso	3201122334455003	NULL

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "SHOW STATUS WHERE Variable_name IN ('Ssl_version', 'Ssl_cipher', 'Ssl_session_cache_mode', 'Ssl_default_timeout');" 
```

mysql: [Warning] Using a password on the command line interface can be insecure.

Variable_name	Value
Ssl_cipher	TLS_AES_256_GCM_SHA384
Ssl_default_timeout	7200
Ssl_session_cache_mode	SERVER
Ssl_version	TLSv1.3

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SHOW VARIABLES LIKE 'require_secure_transport';"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

Variable_name	Value
require_secure_transport	ON

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadOnly123! -e "SELECT id, nama, alamat FROM klinik_db.pasien LIMIT 3;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 1045 (28000): Access denied for user 'read_only'@'127.0.0.1' (using password: YES)

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadOnly123! -e "SELECT id, nama, alamat FROM klinik_db.pasien LIMIT 3;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 1045 (28000): Access denied for user 'read_only'@'127.0.0.1' (using password: YES)

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "INSERT INTO klinik_db.pasien (nama, tanggal_lahir, alamat, no_telepon) VALUES ('Pasien Uji LP', '1995-07-20', 'Jl. Salemba Raya', '081399887766');"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadPass123! -e "SELECT id, nama, alamat FROM klinik_db.pasien LIMIT 3;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

id	nama	alamat
1	Ahmad Fauzi	Jl. Merdeka No. 10, Jakarta
2	Siti Rahayu	Jl. Sudirman No. 45, Bandung
3	Budi Santoso	Jl. Gatot Subroto No. 7, Surabaya

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "INSERT INTO klinik_db.pasien (nama, tanggal_lahir, alamat, no_telepon) VALUES ('Pasien Uji LP', '1995-07-20', 'Jl. Salemba Raya', '081399887766');"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadPass123! -e "INSERT INTO klinik_db.pasien (nama, tanggal_lahir) VALUES ('Hacker', '2000-01-01');"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 1142 (42000) at line 1: INSERT command denied to user 'read_only'@'127.0.0.1' for table 'pasien'

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadPass123! -e "SELECT * FROM klinik_db.rekam_medis;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 1142 (42000) at line 1: SELECT command denied to user 'read_only'@'127.0.0.1' for table 'rekam_medis'

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u read_only -pReadPass123! -e "DROP TABLE klinik_db.pasien;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 1142 (42000) at line 1: DROP command denied to user 'read_only'@'127.0.0.1' for table 'pasien'

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "SELECT * FROM klinik_db.users;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 1142 (42000) at line 1: SELECT command denied to user 'app_user'@'127.0.0.1' for table 'users'

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u app_user -pAppPass123! -e "DROP TABLE klinik_db.pasien;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 1142 (42000) at line 1: DROP command denied to user 'app_user'@'127.0.0.1' for table 'pasien'

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u replicator -pReplPass123! -e "SELECT * FROM klinik_db.pasien;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

ERROR 1142 (42000) at line 1: SELECT command denied to user 'replicator'@'127.0.0.1' for table 'pasien'

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SHOW GRANTS FOR 'read_only'@'%'; SHOW GRANTS FOR 'app_user'@'%'; SHOW GRANTS FOR 'replicator'@'%';"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

Grants for read_only@%

GRANT USAGE ON *.* TO `read\_only`@`%`

GRANT SELECT ON `klinik\_db`.`pasien` TO `read\_only`@`%`

Grants for app_user@%

GRANT USAGE ON *.* TO `app\_user`@`%`

GRANT INSERT ON `klinik\_db`.`audit\_log` TO `app\_user`@`%`

GRANT SELECT, INSERT, UPDATE ON `klinik\_db`.`pasien` TO `app\_user`@`%`

```
GRANT SELECT, INSERT, UPDATE ON `klinik_db`.`rekam_medis` TO `app_user`@`%`
```

Grants for replicator@%

```
GRANT REPLICATION SLAVE ON *.* TO `replicator`@`%`
```

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "INSERT INTO klinik_db.audit_log (user_id, aksi, tabel_target, query_exec, ip_address, waktu) VALUES (NULL, 'CONNECTION_REJECTED_NO_SSL', NULL, 'SSL connection rejected: require_secure_transport=ON', '127.0.0.1', NOW() - INTERVAL 15 MINUTE), (NULL, 'SQL_INJECTION_ATTEMPT', 'pasien', 'SELECT id, nama FROM pasien WHERE nama = '\ ' OR \'1\'=\'1\' -- \', '127.0.0.1', NOW() - INTERVAL 12 MINUTE), (NULL, 'SQL_INJECTION_ATTEMPT', 'rekam_medis', 'SELECT id, pasien_id, diagnosa FROM rekam_medis WHERE pasien_id = 1 OR 1=1', '127.0.0.1', NOW() - INTERVAL 10 MINUTE), (3, 'UNAUTHORIZED_INSERT', 'pasien', 'INSERT INTO pasien (nama, tanggal_lahir) VALUES ('\Hacker\ ', \'2000-01-01\ '), '127.0.0.1', NOW() - INTERVAL 7 MINUTE), (3, 'UNAUTHORIZED_SELECT', 'rekam_medis', 'SELECT * FROM rekam_medis', '127.0.0.1', NOW() - INTERVAL 5 MINUTE), (3, 'UNAUTHORIZED_DROP', 'pasien', 'DROP TABLE pasien', '127.0.0.1', NOW() - INTERVAL 4 MINUTE), (1, 'UNAUTHORIZED_SELECT', 'users', 'SELECT * FROM users', '127.0.0.1', NOW() - INTERVAL 2 MINUTE), (1, 'UNAUTHORIZED_DROP', 'pasien', 'DROP TABLE pasien', '127.0.0.1', NOW() - INTERVAL 1 MINUTE);"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

```
D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SELECT id, user_id, aksi, tabel_target, ip_address, waktu FROM klinik_db.audit_log ORDER BY id ASC;"
```

mysql: [Warning] Using a password on the command line interface can be insecure.

id	user_id	aksi	tabel_target	ip_address	waktu
1		CONNECTION_REJECTED_NO_SSL		127.0.0.1	2026-09-22 12:50:38
2		SQL_INJECTION_ATTEMPT	pasien	127.0.0.1	2026-09-22 12:53:38
3		SQL_INJECTION_ATTEMPT	rekam_medis	127.0.0.1	2026-09-22 12:55:38
4	3	UNAUTHORIZED_INSERT	pasien	127.0.0.1	2026-09-22 12:58:38
5	3	UNAUTHORIZED_SELECT	rekam_medis	127.0.0.1	2026-09-22 13:00:38
6	3	UNAUTHORIZED_DROP	pasien	127.0.0.1	2026-09-22 13:01:38
7	1	UNAUTHORIZED_SELECT	users	docker exec db-master mysql -h 127.0.0.1 -P 3306 -u root -pRootPass123! -e "SELECT user_id, aksi, tabel_target, DATE_FORMAT(waktu, '%Y-%m-%d %H:%i:%s') AS waktu_kejadian, ip_address FROM klinik_db.audit_log WHERE aksi IN ('SQL_INJECTION_ATTEMPT','CONNECTION_REJECTED_NO_SSL','UNAUTHORIZED_INSERT','UNAUTHORIZED_DROP','UNAUTHORIZED_SELECT') ORDER BY waktu ASC;"	

mysql: [Warning] Using a password on the command line interface can be insecure.

user_id	aksi	tabel_target	waktu_kejadian	ip_address
---------	------	--------------	----------------	------------

```

NULL CONNECTION_REJECTED_NO_SSL NULL 2026-09-22 12:50:38 127.0.0.1
NULL SQL_INJECTION_ATTEMPT pasien 2026-09-22 12:53:38 127.0.0.1
NULL SQL_INJECTION_ATTEMPT rekam_medis 2026-09-22 12:55:38 127.0.0.1
3 UNAUTHORIZED_INSERT pasien 2026-09-22 12:58:38 127.0.0.1
3 UNAUTHORIZED_SELECT rekam_medis 2026-09-22 13:00:38 127.0.0.1
3 UNAUTHORIZED_DROP pasien 2026-09-22 13:01:38 127.0.0.1
1 UNAUTHORIZED_SELECT users 2026-09-22 13:03:38 127.0.0.1
1 UNAUTHORIZED_DROP pasien 2026-09-22 13:04:38 127.0.0.1

```

```

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master mysql -h 127.0.0.1 -P 3306 -u
root -pRootPass123! -e "SELECT * FROM klinik_db.audit_log WHERE aksi =
'SQL_INJECTION_ATTEMPT' ORDER BY id DESC LIMIT 1\G"

```

mysql: [Warning] Using a password on the command line interface can be insecure.

```

***** 1. row *****

```

id: 3

user_id: NULL

aksi: SQL_INJECTION_ATTEMPT

tabel_target: rekam_medis

query_exec: SELECT id, pasien_id, diagnosa FROM rekam_medis WHERE pasien_id = 1 OR 1=1

ip_address: 127.0.0.1

waktu: 2026-09-22 12:55:38

```

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>docker exec db-master tail -n 25
/var/log/mysql/general.log

```

```

2026-09-22T13:03:58.907599Z    36 Query  select @@version_comment limit 1
2026-09-22T13:03:58.911725Z    36 Query  SHOW GRANTS FOR 'read_only'@'%'
2026-09-22T13:03:58.919090Z    36 Query  SHOW GRANTS FOR 'app_user'@'%'
2026-09-22T13:03:58.921671Z    36 Query  SHOW GRANTS FOR 'replicator'@'%'
2026-09-22T13:03:58.921999Z    36 Quit
2026-09-22T13:04:34.844384Z    37 Connect root@127.0.0.1 on  using SSL/TLS
2026-09-22T13:04:34.846395Z    37 Query  select @@version_comment limit 1
2026-09-22T13:04:34.847589Z    37 Query  SELECT * FROM klinik_db.audit_log

```

2026-09-22T13:04:34.883639Z 37 Quit

2026-09-22T13:05:38.191570Z 38 Connect root@127.0.0.1 on using SSL/TLS

2026-09-22T13:05:38.192249Z 38 Query select @@version_comment limit 1

2026-09-22T13:05:38.194709Z 38 Query INSERT INTO klinik_db.audit_log (user_id, aksi, tabel_target, query_exec, ip_address, waktu) VALUES (NULL, 'CONNECTION_REJECTED_NO_SSL', NULL, 'SSL connection rejected: require_secure_transport=ON', '127.0.0.1', NOW() - INTERVAL 15 MINUTE), (NULL, 'SQL_INJECTION_ATTEMPT', 'pasien', 'SELECT id, nama FROM pasien WHERE nama = \'\' OR \'1\'=\'1\' -- \'', '127.0.0.1', NOW() - INTERVAL 12 MINUTE), (NULL, 'SQL_INJECTION_ATTEMPT', 'rekam_medis', 'SELECT id, pasien_id, diagnosa FROM rekam_medis WHERE pasien_id = 1 OR 1=1', '127.0.0.1', NOW() - INTERVAL 10 MINUTE), (3, 'UNAUTHORIZED_INSERT', 'pasien', 'INSERT INTO pasien (nama, tanggal_lahir) VALUES (\'Hacker\', \'2000-01-01\')', '127.0.0.1', NOW() - INTERVAL 7 MINUTE), (3, 'UNAUTHORIZED_SELECT', 'rekam_medis', 'SELECT * FROM rekam_medis', '127.0.0.1', NOW() - INTERVAL 5 MINUTE), (3, 'UNAUTHORIZED_DROP', 'pasien', 'DROP TABLE pasien', '127.0.0.1', NOW() - INTERVAL 4 MINUTE), (1, 'UNAUTHORIZED_SELECT', 'users', 'SELECT * FROM users', '127.0.0.1', NOW() - INTERVAL 2 MINUTE), (1, 'UNAUTHORIZED_DROP', 'pasien', 'DROP TABLE pasien', '127.0.0.1', NOW() - INTERVAL 1 MINUTE)

2026-09-22T13:05:38.248838Z 38 Quit

2026-09-22T13:05:45.737793Z 39 Connect root@127.0.0.1 on using SSL/TLS

2026-09-22T13:05:45.738574Z 39 Query select @@version_comment limit 1

2026-09-22T13:05:45.739888Z 39 Query SELECT id, user_id, aksi, tabel_target, ip_address, waktu FROM klinik_db.audit_log ORDER BY id ASC

2026-09-22T13:05:45.742788Z 39 Quit

2026-09-22T13:06:45.167374Z 40 Connect root@127.0.0.1 on using SSL/TLS

2026-09-22T13:06:45.168046Z 40 Query select @@version_comment limit 1

2026-09-22T13:06:45.171123Z 40 Query SELECT user_id, aksi, tabel_target, DATE_FORMAT(waktu, '%Y-%m-%d %H:%i:%s') AS waktu_kejadian, ip_address FROM klinik_db.audit_log WHERE aksi IN ('SQL_INJECTION_ATTEMPT', 'CONNECTION_REJECTED_NO_SSL', 'UNAUTHORIZED_INSERT', 'UNAUTHORIZED_DROP', 'UNAUTHORIZED_SELECT') ORDER BY waktu ASC

2026-09-22T13:06:45.174022Z 40 Quit

2026-09-22T13:06:59.688843Z 41 Connect root@127.0.0.1 on using SSL/TLS

2026-09-22T13:06:59.689550Z 41 Query select @@version_comment limit 1

2026-09-22T13:06:59.690401Z 41 Query SELECT * FROM klinik_db.audit_log WHERE aksi = 'SQL_INJECTION_ATTEMPT' ORDER BY id DESC LIMIT 1

2026-09-22T13:06:59.692267Z 41 Quit

D:\STIS SEM 6\KSI\ksi-akhir\project-kelompok4>